# EV BATTERY HEALTH PREDICTION

Mini Project Report

BCA Sem-III | Course 303-05: Fundamentals of Machine Learning

Student name: _____

Roll number: _____

Faculty guide: _____

Academic year: _____

Dataset: Kaggle Electric Vehicle (EV) Battery Degradation and Charge
Important: Kaggle describes the supplied 10,000 records as synthetic simulations.

Project report generated from the supplied CSV

# CERTIFICATE

This is to certify that _____ (student name),
Roll number _____, has prepared the mini project titled
"EV Battery Health Prediction" as part of BCA Sem-III, Course 303-05:
Fundamentals of Machine Learning.

The project uses a synthetic Kaggle dataset for learning and demonstration.
Faculty approval of the topic and dataset must be confirmed before submission.

Student signature: _____ Date: _____

Faculty signature: _____ Date: _____

Department seal: _____

## PROBLEM STATEMENT

Estimate battery State of Health (SoH, %) from selected vehicle and charging measurements using supervised regression.

## OBJECTIVES

1. Inspect and clean the supplied CSV.
2. Explore its measurements with plots and summary statistics.
3. Fit and compare Linear, Ridge (L2), and Lasso (L1) regression.
4. Evaluate with five-fold cross-validation and held-out test metrics.
5. Save the selected model and demonstrate one held-out prediction.

## DATASET APPROVAL AND LIMITATION

The course instructions require a real-world dataset and faculty approval. The Kaggle author describes this supplied dataset as synthetic simulations. It therefore does not satisfy the real-world-data requirement unless the faculty explicitly approves it for this learning demonstration. Its scores must not be presented as evidence of real battery diagnostic performance.

# DATASET DESCRIPTION

| Column | Meaning | Unit or kind |
|---|---|---|
| Vehicle_ID | Vehicle record identifier | ID |
| Car_Model | Vehicle model name | Category |
| Battery_Type | Battery chemistry (LFP or NMC) | Category |
| Battery_Capacity_kWh | Listed battery capacity | kWh |
| Vehicle_Age_Months | Vehicle age | months |
| Total_Charging_Cycles | Cumulative charge/discharge count | cycles |
| Avg_Temperature_C | Average operating temperature | °C |
| Fast_Charge_Ratio | Share of fast-charging sessions | 0 to 1, unitless |
| Avg_Discharge_Rate_C | Average discharge rate | C-rate |
| Driving_Style | Driving-style label | Category |
| Internal_Resistance_Ohm | Battery internal resistance | ohms (Ω) |
| SoH_Percent | State of Health; prediction target | % |
| Battery_Status | Health-status label | Category |

Dataset shape: 10000 rows × 13 columns. Model target: SoH_Percent (%).

# DATA PREPROCESSING

| Check | Before cleaning | After cleaning |
|---|---|---|
| Rows (records) | 10000 | 10000 |
| Missing cells (values) | 0 | 0 |
| Duplicate rows (rows) | 0 | 0 |
| IQR outlier values in five input measurements (values) | 101 | 0 |

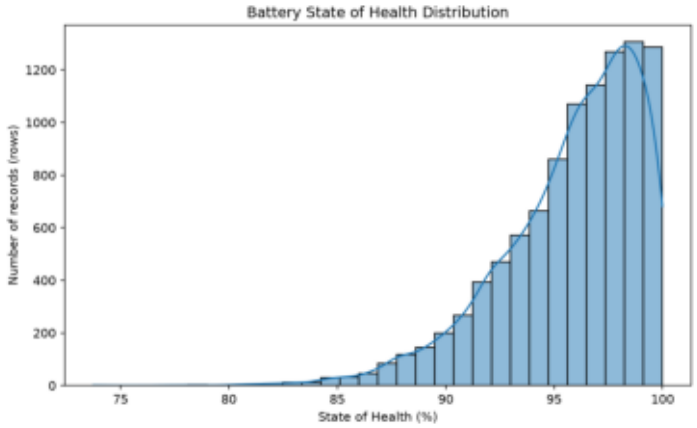
## Cleaning decisions

- Exact duplicate rows were removed.
- Missing numeric values use the median; missing text values use the mode.
- IQR limits were applied to five model-input measurements only.
- SoH target values were not capped; all remaining records were retained.

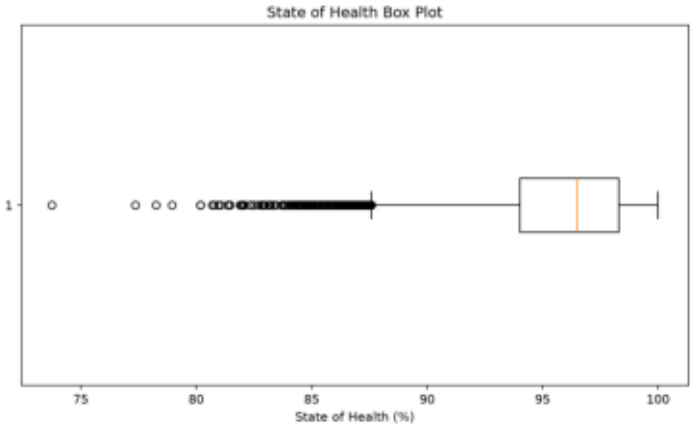
Selected inputs: vehicle age (months), charging cycles (cycles), average temperature (°C), fast-charge ratio (unitless), and discharge rate (C-rate). Excluded: identifier, text categories, Battery_Status, internal resistance, and capacity (possible target leakage or direct target association).

# EXPLORATORY DATA ANALYSIS

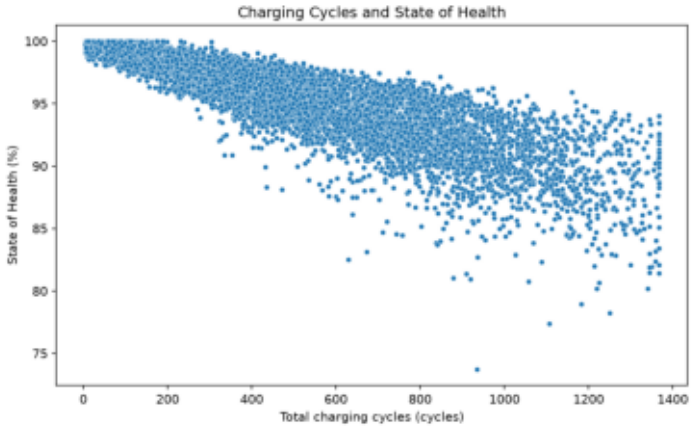
SoH distribution (%)



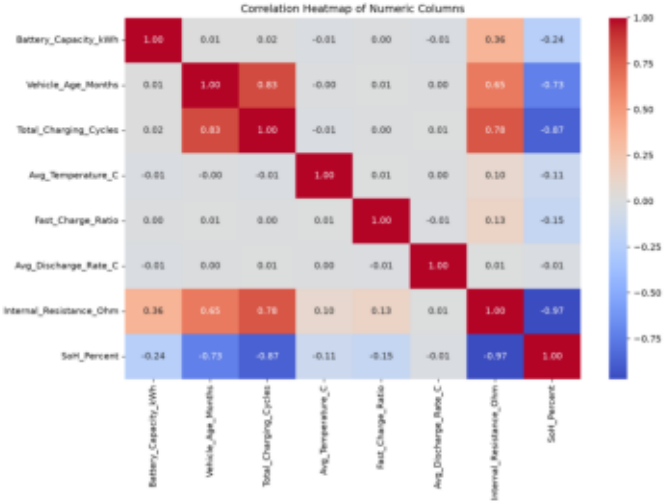

SoH box plot (%)



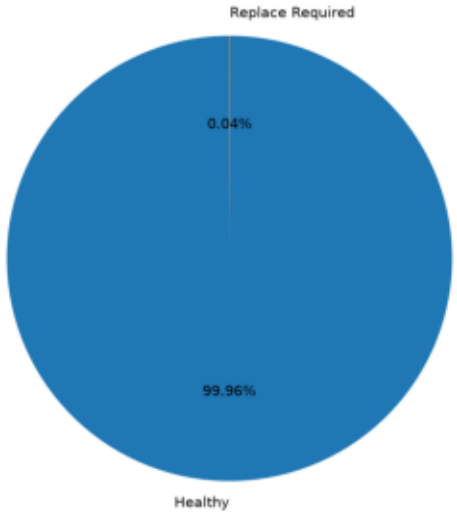

Cycles and SoH (cycles, %)



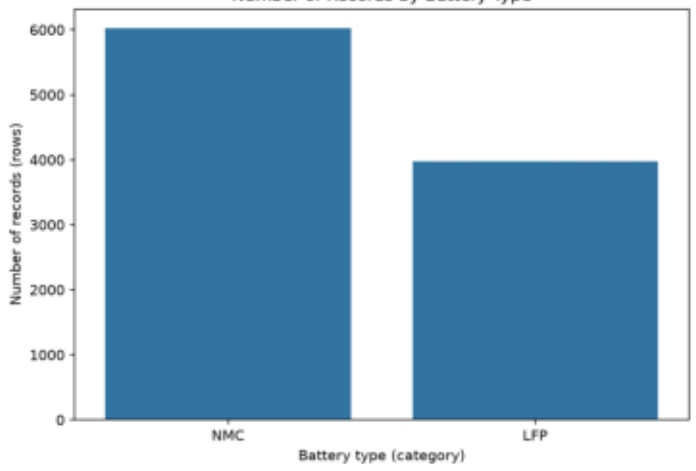

Numeric-column correlation (unitless)




Battery status share (%)




Records by battery type (rows)

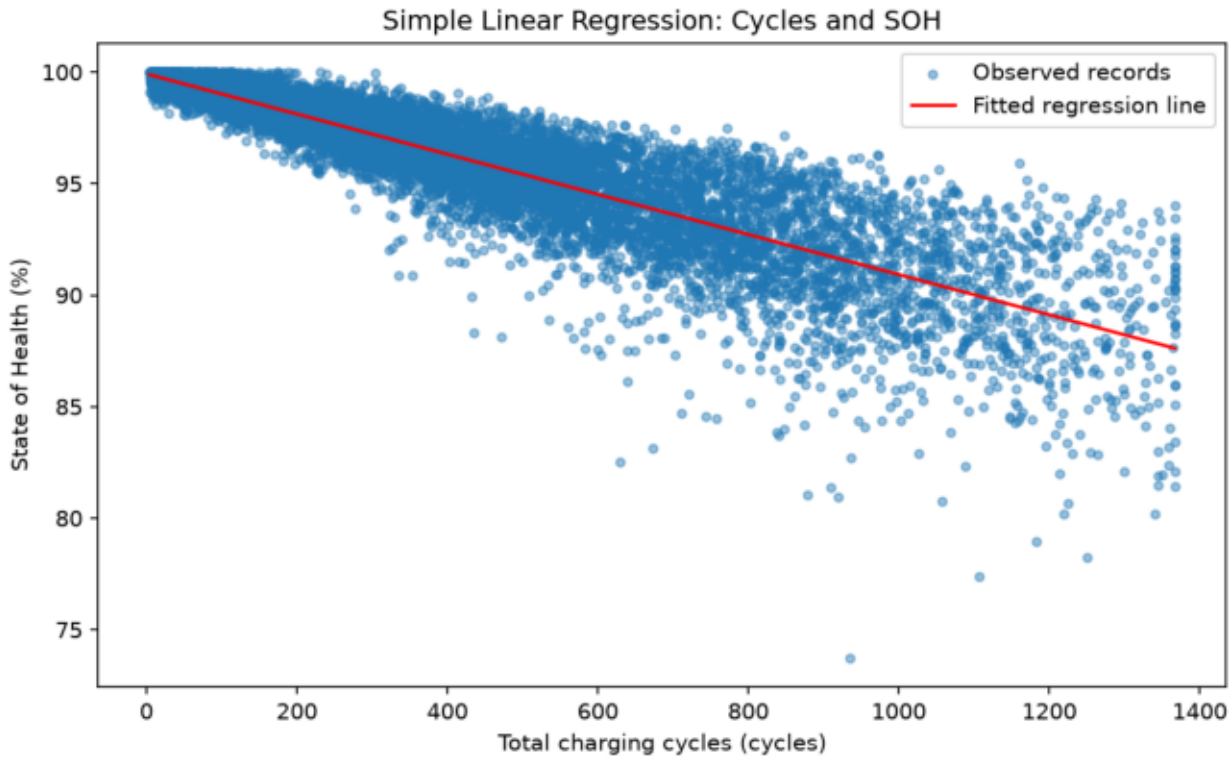


# SIMPLE LINEAR REGRESSION

SOH (%) = -0.009001 (% per cycle) × cycles + 99.920179 (%)

R-squared (unitless): 0.762814

Residual sum of squares (%²): 25094.825246




This one-feature fit is for teaching y = mx + b. Its R-squared is calculated on the cleaned data used to draw the illustration, not the held-out test set.

# MODEL BUILDING AND EVALUATION

Training rows: 8000 | Held-out test rows: 2000
A StandardScaler is fitted inside each training fold. Five-fold cross-validation compares Linear Regression, Ridge (L2), and Lasso (L1).

| Model | Mean CV R2 (unitless) | MAE (%) | MSE (% squared) | RMSE (%) | Test R2 (unitless) |
|---|---|---|---|---|---|
| Linear | 0.8004 | 1.0687 | 2.2163 | 1.4887 | 0.7916 |
| Ridge (L2) | 0.8004 | 1.0687 | 2.2163 | 1.4887 | 0.7916 |
| Lasso (L1) | 0.7976 | 1.0623 | 2.2665 | 1.5055 | 0.7868 |

Selected by mean five-fold CV $R^2$: Linear
MAE and RMSE are in SoH percentage points (%); MSE is in %²; $R^2$ is unitless. Linear and Ridge can tie at the displayed precision. Scores describe only performance on this synthetic dataset and split.

Model-building code pattern:

```
X = df[feature_columns]
y = df['SoH_Percent']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20)
model = make_pipeline(StandardScaler(), LinearRegression())
```

# RESULTS, CONCLUSION, AND FUTURE SCOPE

Best model: Linear

Held-out sample actual SoH: 97.31 %

Held-out sample predicted SoH: 95.41 %

The saved estimator is Python_Code/best_model.pkl.

## Conclusion

The project demonstrates a complete beginner regression workflow: inspection, cleaning, visual EDA, model training, cross-validation, test evaluation, and saving a fitted estimator. Linear and Ridge performed similarly in this run. The synthetic data prevents a real-world battery-health conclusion.

## Future scope

Obtain measured battery test records and faculty approval; validate on vehicles or batteries not seen during training; document units and collection methods; check leakage and generalization; and compare predictions with a suitable battery-health measurement before considering practical use.

Dataset source: Kaggle, Electric Vehicle (EV) Battery Degradation and Charge, https://www.kaggle.com/datasets/bertnardomariouskono/electric-vehicle-ev-battery-degradation-and-charge
The Kaggle author describes the supplied data as synthetic simulations.